

# Notes: Antràs (AER P&P, 2005)

## Property Rights and the International Organization of Production

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# 1 Big picture

**One-sentence summary.** This note summarizes Antràs (2005), which explains how incomplete contracts and residual control rights can be embedded into international trade models to study firms’ joint decisions over where to locate production and whether to organize production inside firm boundaries or through arm’s-length outsourcing.

**Core question.** When firms split production across countries, why do some transactions occur within multinational firm boundaries while others occur through independent suppliers? The paper asks how the “make-or-buy” decision interacts with the “where-to-produce” decision in global production.

**Key answer.** Under incomplete contracts, ownership matters because it changes each party’s ex post bargaining position and therefore each party’s ex ante investment incentives. Integration is more attractive when headquarter services or capital-intensive investments are especially important; outsourcing is more attractive when supplier-side manufacturing investment is more important. When the model is put into international trade settings, it predicts systematic variation in intrafirm trade across industries, countries, firms, and product-cycle stages.

**Why this paper matters.** The paper is a compact overview of the Antràs property-rights approach to multinational firms. It shifts the theory of multinationals away from purely technological explanations, such as scale economies or transport costs, and toward an organizational explanation based on incomplete contracts, asset ownership, residual rights of control, and relationship-specific investment.

**Main objects in the model.**

- $H$ : the final-good producer or headquarters, who supplies the headquarter input  $h$ .
- $M$ : the manufacturing-plant operator, who supplies the component input  $m$ .
- $k \in \{V, O\}$ : ownership structure, where  $V$  denotes vertical integration and  $O$  denotes outsourcing.
- $\ell \in L$ : locational choice, such as domestic production, foreign outsourcing, or foreign integration.
- $\eta$ : headquarter intensity, or the technological importance of input  $h$  relative to input  $m$ .

## 2 Incremental contribution of the paper

**Relative to earlier multinational-firm theories.** Traditional theories of the multinational firm often explain FDI through technological considerations, such as firm-level scale economies, transport costs, or knowledge-transfer costs. Antràs instead gives an organizational theory: multinational production is governed by the allocation of residual rights when contracts are incomplete.

**Relative to Grossman-Hart property-rights theory.** Grossman and Hart (1986) explain firm boundaries using asset ownership and relationship-specific investment. Antràs takes this logic into international trade: the same property-rights mechanism that explains vertical integration inside a country can also explain intrafirm trade, foreign direct investment, international outsourcing, and product-cycle dynamics.

**Relative to trade models of outsourcing.** Earlier outsourcing models often emphasized search costs, information problems, or knowledge dissipation. This paper shows that even without those

frictions, incomplete contracts alone can generate a rich theory of global sourcing and intrafirm trade.

**Relative to empirical regularities.** The framework rationalizes two broad facts: intrafirm import shares are higher in more capital-intensive or headquarter-intensive industries, and intrafirm import shares are higher when exports come from more capital-abundant countries. With firm heterogeneity, it also explains why different organizational forms coexist within the same industry.

**Methodological contribution.** The paper shows how to combine the property-rights approach with the apparatus of international trade theory. This produces a unified framework in which ownership, location, productivity heterogeneity, country factor prices, and contractual institutions jointly determine the international organization of production.

### 3 The basic partial-equilibrium model

#### 3.1 Demand, production, and revenue

A final-good producer faces an isoelastic demand curve. A convenient representation is

$$y = \lambda p^{-1/(1-\alpha)}, \quad 0 < \alpha < 1,$$

where  $y$  is quantity,  $p$  is price, and  $\lambda$  is a demand shifter. The inverse demand curve is therefore

$$p = \lambda^{1-\alpha} y^{-(1-\alpha)},$$

so revenue is

$$py = \lambda^{1-\alpha} y^\alpha.$$

Production uses two specialized inputs:

$$y = \left(\frac{h}{\eta}\right)^\eta \left(\frac{m}{1-\eta}\right)^{1-\eta}, \quad 0 < \eta < 1.$$

The parameter  $\eta$  measures the importance of headquarter services  $h$ . A higher  $\eta$  means that the final-good producer's investment is more important for revenue generation.

Combining demand and technology gives the revenue function

$$R(h, m) = \lambda^{1-\alpha} \left[ \left(\frac{h}{\eta}\right)^\eta \left(\frac{m}{1-\eta}\right)^{1-\eta} \right]^\alpha.$$

Equivalently,

$$R(h, m) = Ah^{\alpha\eta} m^{\alpha(1-\eta)},$$

where

$$A = \lambda^{1-\alpha} \eta^{-\alpha\eta} (1-\eta)^{-\alpha(1-\eta)}.$$

Since  $\alpha < 1$ , revenue is concave in the pair of investments  $(h, m)$ .

**Economic message.** The two inputs are relationship-specific and complementary. A failure by either party to invest reduces the total surplus. The ownership structure matters because it determines how much of the return to investment each party expects to capture.

### 3.2 Incomplete contracts and ex post bargaining

Contracts are incomplete. Ex ante, the parties can contract on the ownership structure and a lump-sum transfer, but they cannot write an enforceable contract specifying the exact type, quality, and price of the specialized inputs. After  $h$  and  $m$  are produced, the parties bargain over the relationship surplus.

Under outsourcing, neither party owns the other's input. If bargaining breaks down, the specialized inputs are useless outside the relationship, so both outside options are zero. If  $H$  has Nash bargaining weight  $\beta$ , then under outsourcing

$$\beta_O = \beta.$$

Under integration,  $H$  owns the manufacturing plant. If bargaining breaks down,  $H$  can fire  $M$  and seize the already produced  $m$ , but can use it only imperfectly. Let  $\delta \in (0, 1)$  measure how effective production is without  $M$ 's cooperation. Because revenue is proportional to output to the power  $\alpha$ ,  $H$ 's outside option under integration is proportional to  $\delta^\alpha R(h, m)$ . The final producer's effective ex post revenue share becomes

$$\beta_V = \delta^\alpha + \beta(1 - \delta^\alpha) = \beta + (1 - \beta)\delta^\alpha > \beta_O.$$

Therefore integration shifts ex post surplus toward  $H$ .

**Economic message.** Integration does not eliminate ex post bargaining. It changes the outside option of the headquarters and therefore changes the ex post division of surplus. This is the property-rights mechanism.

### 3.3 Investment incentives under each ownership structure

For a given ownership structure  $k \in \{V, O\}$ , the revenue share of  $H$  is  $\beta_k$ . The final-good producer chooses  $h$  to maximize its private return,

$$h_k = \arg \max_h \{ \beta_k R(h, m_k) - c_h h \},$$

while the manufacturing operator chooses  $m$  to maximize

$$m_k = \arg \max_m \{ (1 - \beta_k) R(h_k, m) - c_m m \}.$$

The first-order conditions are

$$\beta_k R_h(h_k, m_k) = c_h, \quad (1 - \beta_k) R_m(h_k, m_k) = c_m.$$

Using the Cobb-Douglas revenue function,

$$R_h = \frac{\alpha \eta R}{h}, \quad R_m = \frac{\alpha(1 - \eta) R}{m}.$$

Thus the incentive-compatible investment levels satisfy

$$\begin{aligned} c_h h_k &= \alpha \eta \beta_k R(h_k, m_k), \\ c_m m_k &= \alpha(1 - \eta)(1 - \beta_k) R(h_k, m_k). \end{aligned}$$

These expressions summarize the central distortion:

- $H$  invests as if it receives only  $\beta_k$  of the marginal revenue from  $h$ .
- $M$  invests as if it receives only  $1 - \beta_k$  of the marginal revenue from  $m$ .
- Both parties underinvest relative to first best because neither captures the full marginal product of investment.

### 3.4 Operating surplus and the role of ownership

Using the first-order conditions, variable operating profit under ownership structure  $k$  is

$$\Pi_k^{\text{var}} = R(h_k, m_k) - c_h h_k - c_m m_k.$$

Substituting the private-investment conditions gives

$$\Pi_k^{\text{var}} = [1 - \alpha\eta\beta_k - \alpha(1 - \eta)(1 - \beta_k)] R(h_k, m_k).$$

The level of  $R(h_k, m_k)$  itself depends on  $\beta_k$ , because  $\beta_k$  governs how much each side invests. Combining the two first-order conditions with the revenue function yields

$$R(h_k, m_k)^{1-\alpha} = A \left( \frac{\alpha\eta\beta_k}{c_h} \right)^{\alpha\eta} \left( \frac{\alpha(1 - \eta)(1 - \beta_k)}{c_m} \right)^{\alpha(1-\eta)}.$$

Hence, up to common terms, the organizational efficiency of a revenue share  $\beta_k$  can be summarized by

$$\Theta(\beta_k; \eta) = (\beta_k)^{\frac{\alpha\eta}{1-\alpha}} (1 - \beta_k)^{\frac{\alpha(1-\eta)}{1-\alpha}} [1 - \alpha\eta\beta_k - \alpha(1 - \eta)(1 - \beta_k)].$$

The firm chooses vertical integration when the value associated with  $\beta_V$  exceeds the value associated with  $\beta_O$  after accounting for fixed costs and transfers.

**Interpretation.** Raising  $\beta_k$  increases  $H$ 's incentive to invest in headquarter services but reduces  $M$ 's incentive to invest in manufacturing components. Therefore the optimal ownership structure depends on whether  $h$  or  $m$  is more important for production.

### 3.5 The integration threshold

Because  $\beta_V > \beta_O$ , integration gives more bargaining power to  $H$ . This is valuable when  $H$ 's input  $h$  is especially important. Outsourcing gives relatively more incentives to  $M$ , which is valuable when  $m$  is especially important.

The central result is therefore a threshold rule:

$$\exists \hat{\eta} \in (0, 1) \quad \text{such that} \quad \begin{cases} \eta > \hat{\eta} & \Rightarrow \text{integration dominates outsourcing,} \\ \eta < \hat{\eta} & \Rightarrow \text{outsourcing dominates integration.} \end{cases}$$

**Economic message.** The boundary of the firm is determined by the relative importance of relationship-specific investments. Residual rights should be assigned to the party whose investment contributes more to the relationship surplus.

## 4 Adding location: ownership and geography

The two-country version adds a locational choice  $\ell \in L$ . Different locations can imply different input costs, fixed costs, outside options, bargaining parameters, and legal institutions. The firm now chooses both ownership  $k \in \{V, O\}$  and location  $\ell$  to solve

$$\max_{k \in \{V, O\}, \ell \in L} \pi_k^\ell = R^\ell(h_k^\ell, m_k^\ell) - c_h^\ell h_k^\ell - c_m^\ell m_k^\ell - f_k^\ell g^\ell(c_h^\ell, c_m^\ell) - \bar{U}^\ell,$$

subject to

$$\begin{aligned} h_k^\ell &= \arg \max_h \left\{ \beta_k^\ell R^\ell(h, m_k^\ell) - c_h^\ell h \right\}, \\ m_k^\ell &= \arg \max_m \left\{ (1 - \beta_k^\ell) R^\ell(h_k^\ell, m) - c_m^\ell m \right\}. \end{aligned}$$

This program is the common backbone of the three applications in the paper:

- a world equilibrium with homogeneous firms;
- a world equilibrium with heterogeneous firms;
- a product-cycle model with weaker contract enforceability across borders.

**Key idea.** Location affects production costs and contractual frictions. Ownership affects investment incentives. Global sourcing is therefore a joint solution to cost minimization and incentive provision.

## 5 Application I: homogeneous firms and intrafirm trade patterns

### 5.1 Setup

In the homogeneous-firm version, there are two differentiated sectors,  $Y$  and  $Z$ . Preferences are Dixit-Stiglitz within sectors, and final-good producers in the North source a composite intermediate input that combines  $h$  and  $m$ . The headquarter intensity is higher in sector  $Y$  than in sector  $Z$ :

$$\eta_Y > \eta_Z.$$

The input  $h$  is assumed to be capital-intensive and the input  $m$  is labor-intensive. In an extreme-factor-intensity simplification,

$$c_h^\ell = r^\ell, \quad c_m^\ell = w^\ell,$$

where  $r^\ell$  is the rental rate of capital and  $w^\ell$  is the wage in location  $\ell$ .

The location decision reduces to choosing whether  $h$  and  $m$  are combined in the North or the South. The ownership decision can be analyzed separately from location under the simplifying assumptions of the model.

### 5.2 Industry-level prediction

If

$$\eta_Y > \hat{\eta} > \eta_Z,$$

then sector  $Y$  uses vertical integration while sector  $Z$  uses outsourcing. Since sector  $Y$  is more headquarter- or capital-intensive, the model predicts that intrafirm trade shares should be larger in more capital-intensive industries.

**Prediction 1.** Across industries, the share of intrafirm imports in total imports is higher in industries that are more capital intensive or headquarter intensive.

### 5.3 Country-level prediction

If the South is capital-scarce relative to the North, then the South produces a disproportionate share of the more labor-intensive sector. A more capital-abundant Southern country will export relatively more of the capital-intensive sector  $Y$ . Since sector  $Y$  is the sector associated with intrafirm trade, the model predicts a positive relationship between the exporting country's capital-labor ratio and the intrafirm-import share.

**Prediction 2.** Across exporting countries, the share of intrafirm imports in total U.S. imports is higher when the exporting country is more capital abundant.

**Economic message.** Intrafirm trade is not random across industries and countries. It is systematically linked to factor intensities because factor intensities determine the relative importance of the headquarter input and therefore the optimal ownership structure.

## 6 Application II: heterogeneous firms and global sourcing

### 6.1 Why firm heterogeneity matters

The homogeneous-firm model predicts corner outcomes: within an industry, all firms should choose the same ownership structure. In the data, however, intrafirm import shares vary smoothly across industries, and multiple organizational forms coexist within industries. This motivates the heterogeneous-firm extension in Antràs and Helpman (2004).

The extension adds Melitz-type productivity heterogeneity. Firms draw productivity  $\theta$  from a Pareto distribution,

$$G(\theta) = 1 - \left(\frac{b}{\theta}\right)^z, \quad \theta \geq b > 0,$$

where lower  $z$  means greater productivity dispersion.

### 6.2 Location and fixed-cost ranking

The final good and headquarter services are produced in the North, while manufactured components can be produced in the North or in the South. The South has lower variable costs:

$$w_N > w_S.$$

However, foreign sourcing and integration require higher fixed costs. A natural ranking is

$$f_V^S > f_O^S > f_V^N > f_O^N.$$

This ranking means:



- foreign sourcing is fixed-costly relative to domestic sourcing;
- integration is fixed-costly relative to outsourcing within a given location;
- foreign integration is the most fixed-cost-intensive organizational form;
- domestic outsourcing is the least fixed-cost-intensive organizational form.

### 6.3 Sorting by productivity

Because foreign sourcing has lower marginal costs but higher fixed costs, only sufficiently productive firms find it profitable to source abroad. Because integration improves investment incentives in headquarter-intensive sectors but has higher fixed costs, the most productive firms are also more likely to internalize.

In the benchmark headquarter-intensive sector, firms sort as follows:

$$\begin{aligned} \text{exit} &\rightarrow \text{domestic outsourcing} \rightarrow \text{domestic integration} \\ &\rightarrow \text{foreign outsourcing} \rightarrow \text{foreign integration.} \end{aligned}$$

The ranking is in increasing productivity. The most productive firms can afford the fixed costs of foreign integration and benefit most from low foreign variable costs and improved headquarter incentives. Less productive surviving firms choose domestic forms, and the least productive exit.

**Economic message.** Global sourcing is a selection problem. More productive firms choose organizational forms with higher fixed costs and lower variable costs. Therefore firm productivity jointly predicts internationalization and internalization.

### 6.4 Comparative statics

The heterogeneous-firm model generates richer empirical predictions:

- **Headquarter intensity.** The share of intrafirm imports in total component imports is higher in industries with higher  $\eta$ .
- **Productivity dispersion.** The intrafirm-import share is higher in industries with greater productivity dispersion, because more mass lies in the high-productivity tail that can support high-fixed-cost integration.
- **Transport costs.** Higher transport costs can increase the intrafirm share among importers because only the most productive firms continue foreign sourcing, and these firms are more likely to integrate.
- **North-South wage gaps.** Larger wage differences increase international sourcing, but they can also increase the relative prevalence of foreign outsourcing among foreign sourcing firms.
- **Falling trade costs.** Lower trade costs increase international sourcing and can change the composition of sourcing toward more outsourcing.

## 7 Application III: contract enforceability and the product cycle

### 7.1 Motivation

The previous sections assumed that contract incompleteness is the same for domestic and foreign transactions. Antràs (2005) relaxes this assumption. Domestic contracts are assumed to be more enforceable than international contracts. This matters because offshoring creates a trade-off between lower foreign wages and worse contractual enforcement.

### 7.2 Domestic complete contracts versus foreign incomplete contracts

Suppose  $h$  and final assembly remain in the North, while the firm chooses whether to source component  $m$  domestically or from the South. Domestic contracts are complete, so domestic sourcing avoids the hold-up problem. Foreign contracts are incomplete, so foreign sourcing creates investment distortions.

When the North-South wage gap  $w_N/w_S$  is sufficiently high, there are two threshold levels of headquarter intensity,

$$\underline{\eta} < \bar{\eta},$$

with the following organizational pattern:

$$\begin{cases} \eta > \bar{\eta} & \Rightarrow \text{produce } m \text{ in the North,} \\ \underline{\eta} < \eta < \bar{\eta} & \Rightarrow \text{source } m \text{ from an integrated supplier in the South,} \\ \eta < \underline{\eta} & \Rightarrow \text{source } m \text{ from a nonintegrated supplier in the South.} \end{cases}$$

#### Interpretation.

- When  $\eta$  is high, headquarter investment is too important to expose the relationship to foreign contractual frictions, so domestic sourcing is optimal.
- When  $\eta$  is intermediate, the labor-cost advantage of the South is attractive, but  $H$  still needs control rights to protect headquarter incentives, so foreign integration is optimal.
- When  $\eta$  is low, manufacturing investment is relatively important, so foreign outsourcing is optimal.

### 7.3 Product-cycle implication

A product becomes more standardized over its life cycle. In the model, standardization is represented by a gradual decline in  $\eta$ : headquarter services, product development, and marketing become less important relative to manufacturing and assembly.

This generates a three-stage product cycle:

$$\text{domestic production} \rightarrow \text{foreign integration / FDI} \rightarrow \text{foreign outsourcing.}$$

**Economic message.** The location of production and the boundary of the firm evolve together over the product life cycle. New products stay close to headquarters because contracts are incomplete and headquarter services are important. Mature products can be moved abroad, first inside firm boundaries and later to independent suppliers.

## 8 How to read the paper's main mechanism

### 8.1 The central trade-off

The whole paper can be read as a repeated application of one trade-off:

cost advantage of decentralization/offshoring  
versus  
incentive advantage of control/domestic enforcement.

Ownership shifts ex post bargaining power. Location shifts input costs and contractual enforceability. Productivity determines whether the firm can pay the fixed costs associated with more complex organizational forms.

### 8.2 Why integration is not always optimal

Integration gives more residual control to  $H$ . This improves  $H$ 's investment incentives but weakens  $M$ 's incentives. Because  $m$  is relationship-specific and noncontractible, reducing  $M$ 's incentive can be costly. Therefore integration is optimal only when the marginal contribution of  $H$ 's investment is sufficiently high.

### 8.3 Why outsourcing is not simply market exchange

Outsourcing is not a frictionless spot-market transaction. The supplier must make relationship-specific investments, and the parties bargain after investments are sunk. Arm's-length sourcing is therefore still subject to hold-up. The key difference from integration is the allocation of residual rights and the resulting bargaining shares.

### 8.4 Why global sourcing is more than a labor-cost story

The South has lower wages, but foreign sourcing may involve worse contracting and higher fixed organizational costs. The firm does not offshore all production stages simply because wages are lower. It offshores when the wage advantage is large enough relative to the incentive losses from incomplete contracts.

## 9 Relation to the user's research interests and broader research agenda

This paper is useful for thinking about modern finance, trade, and organization research because it turns a broad globalization phenomenon into an allocation-of-control problem. The key research logic is:

institutions and contracts  $\rightarrow$  organizational form  
 $\rightarrow$  investment incentives  
 $\rightarrow$  trade and production patterns.

For research on firms, banking, or financial intermediation, the paper offers a transferable structure. Many economic outcomes are not determined only by prices or technology, but by who controls assets, who bears contractual risk, and who has incentives to invest when future contingencies cannot be fully contracted.

In the context of trade shocks or tariff shocks, the paper suggests a way to think about organizational responses. If a shock changes the relative attractiveness of domestic sourcing, foreign outsourcing, or foreign integration, then firms may reorganize production not only because of cost changes but also because contractual and control problems change. This provides a bridge between trade, corporate organization, and investment incentives.

## 10 Short summary

**Research question.** How do property rights and incomplete contracts shape the international organization of production?

**Model ingredients.** Production requires two relationship-specific inputs: headquarter services  $h$  and manufacturing components  $m$ . Contracts are incomplete, so ownership affects ex post bargaining and ex ante investment incentives.

**Core mechanism.** Integration raises the headquarters' bargaining share and encourages investment in  $h$ , but it weakens the supplier's incentives to invest in  $m$ . Outsourcing does the reverse. The efficient boundary of the firm depends on whether production is more intensive in headquarter services or manufacturing components.

**Main predictions.**

- Intrafirm trade is more common in headquarter- or capital-intensive industries.
- Intrafirm imports are more common from capital-abundant countries.
- With heterogeneous firms, the most productive firms choose the highest-fixed-cost organizational forms, including foreign integration.
- Along the product cycle, production moves from domestic production to foreign integration and finally to foreign outsourcing as goods become standardized.

**Takeaway.** The multinational firm is not only a technological or geographic object; it is an organizational response to incomplete contracts. Global production patterns reflect how firms allocate residual control rights to protect relationship-specific investments across borders.

## 11 Conclusion

Antràs (2005) provides a concise synthesis of a property-rights approach to multinational production. The central insight is that international production requires not only choosing locations but also assigning control rights. Because contracts are incomplete, ownership affects bargaining power, bargaining power affects investment incentives, and investment incentives affect the profitability of different global sourcing modes.

The paper's framework explains why intrafirm trade is systematically related to industry factor

intensity, country factor abundance, firm productivity, trade costs, and product maturity. It also shows why multinational production cannot be understood purely as a response to low wages abroad. Offshoring and integration are jointly shaped by production costs, fixed organizational costs, and the quality of contract enforcement.

**Final takeaway.** A firm chooses integration when headquarter incentives are sufficiently important; it chooses outsourcing when supplier incentives dominate; and it chooses foreign production only when lower foreign costs compensate for additional fixed costs and contractual distortions. This is the organizing logic behind the property-rights theory of global production.